

Scenario Demonstration 1: Data Preparation

Independent Task Sheet | Phase: Actual | SD1

Organisation: Apex Sports Academy | **Analyst:** Jamie | **Period:** January – December 2024

Business Question

Which session types, facilities and membership tiers generate the highest bookings and attendance rates, and what patterns suggest members are at risk of disengaging?

Tasks

1. Import and Inspect

- Open SD1_START.xlsx. Inspect all Power Query connections.
- Record row counts for each source: Session Bookings, Members, and Lookup tables.
- Note any data quality issues you can see before making changes.

2. Clean the Data in Power Query

- Standardise text casing in all relevant columns.
- Trim and clean leading/trailing spaces from all text fields.
- Replace null attendance_status values with an appropriate label. Document your reasoning.
- Address null fee_paid values. Document each decision.
- Remove duplicate records. Record how many were removed and from which table.

3. Validate Keys and Merge Datasets

- Verify that booking_id is a valid primary key after cleaning.
- Merge Session Bookings with Members using the correct join type. Justify your choice.
- Merge with Facility Lookup and Session Type Lookup.
- Verify row counts after each merge. Document any unmatched records.

4. Add Calculated Columns

- Add booking_month (name of month).
- Add quarter (Q1, Q2, Q3, Q4).
- Add attended_flag (Yes / No).
- Add fee_category (Free/Cancelled, Low, Standard, Premium).
- Add member_full_name (first_name + last_name).

5. Validate and Load

- Complete all validation checks. Document results in a Validation_Summary sheet.
- Complete a Cleaning_Log recording every cleaning decision made.
- Load the final clean dataset to Excel as a formatted table.
- Save as SD1_COMPLETE.xlsx.

Required Deliverables

• SD1_COMPLETE.xlsx — clean, merged dataset (122 rows, 20 columns).
• Validation_Summary sheet — row counts, null checks, duplicate checks.
• Cleaning_Log sheet — all decisions documented with reasoning.
• Power Query Applied Steps intact and reviewable.

KSB Mapping: S1, S2, S3, S4, S5 | K7, K8, K10, K11 | B2, B3, B4

Level 3 Data Technician Apprenticeship — SD1
Actual